

Satyam Awasthi

Researcher & Software Engineer

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Professional Summary

Computer Science researcher specializing in the intersection of **Autonomous Decision-Making**, **Spatial Perception**, and **Robot Learning** within Embodied Systems, with a core focus on engineering verifiably safe, constraint-bound execution loops.

Education

University of California, Santa Barbara (UCSB)

Santa Barbara, CA

M.S. in Computer Science (GPA: 3.91/4.00)

Sep 2021 – Jun 2023

Graduate Research at the Four Eyes Lab focusing on calibrating eye-tracking frameworks for mixed-reality locomotion and multi-modal spatial perception.

Advisor: [Prof. Tobias Höllerer](#) · Co-Advisor: [Prof. Michael Beyeler](#)

Indian Institute of Technology, Kharagpur (IIT Kharagpur)

Kharagpur, India

B.Tech. in Electrical Engineering, Minor in Computer Science (GPA: 8.98/10.0)

Jul 2016 – Jul 2020

Research focused on 2D kinematic profiles, decentralized multi-agent motion planning, and hardware-agnostic stacks for Sim2Real transfer.

Skills

Programming & Simulation: Python, C++, MATLAB, Kotlin, Java, Gazebo, Rviz, Unity, Git

Robotics, Vision & AI Frameworks: ROS/ROS2, OpenCV, PyTorch, LangChain, Pydantic, ARCore/iOS ARKit

Core Methodologies: State Estimation (EKF), Control Theory (MPC), Kinematic Modeling, Spatial Perception

Fields of Interest: Autonomous Decision-Making, Spatial Perception, Robot Learning

Publications

Refereed Conference Papers

SmartWatch for Wall Writing: Real-time Transcription of Wall Writing from Inertial Sensing

Snigdha Das, **Satyam Awasthi**, Abdul Shamnar P, Pradipta De, Sandip Chakraborty, and Bivas Mitra.

Proceedings of the 14th International Conference on COMmunication Systems & NETworkS (COMSNETS), 2022.

DOI: [10.1109/COMSNETS53615.2022.9668531](https://doi.org/10.1109/COMSNETS53615.2022.9668531)

Engineered an adaptive coordinate transformation pipeline using gravitational alignment and PUPD micro-gesture filtering to resolve drift-free, 2D kinematic profiles of wall-writing from wearable inertial sensors.

Refereed Poster Papers

Eye Tracking Performance in Mobile Mixed Reality

Satyam Awasthi, Vivian Ross, Sydney Lim, Michael Beyeler, and Tobias Höllerer.

IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), 2024.

DOI: [10.1109/VRW62533.2024.00321](https://doi.org/10.1109/VRW62533.2024.00321)

Expanded the tracking framework to benchmark low-latency performance profiles and gaze errors via a robust 3-device user study across diverse mixed-reality locomotion layouts. Supported by NSF Grant #2211784.

EyeTTS: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion

Satyam Awasthi, Vivian Ross, Michael Beyeler, and Tobias Höllerer.

IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct), 2023.

DOI: [10.1109/ISMAR-Adjunct60411.2023.00104](https://doi.org/10.1109/ISMAR-Adjunct60411.2023.00104)

Developed a novel [eye-tracking test suite \(EyeTTS\)](#) and engineered unique software calibration pipelines optimized to correct tracking drift during active human locomotion; validated via a 2-device user study. Supported by NSF Grant #2211784.

Experience

Intuit Inc

Mountain View, CA · Mar 2024 – Jan 2026

Software Engineer 2

- Built **dual-stage agentic decision pipelines** using tool-schema routing for real-time intent classification and sequential state tracking.
- Developed a decoupled **generative UI engine** leveraging RAG and structured LLMs to optimize deterministic component layouts.

Yahoo Inc

Mountain View, CA · Apr 2023 – Nov 2023

Software Dev Engineer II (IC3)

- Developed **client-side rule engines** for asynchronous receipt payloads, executing deterministic filtering for **real-time decision support**.
- Designed a dynamic, state-driven carousel UI framework optimized for client-side rendering efficiency.

Coupang Global LLC

Mountain View, CA · Jun 2022 – Sep 2022

Software Engineering Intern

- Engineered **context-aware heuristics** to prune multi-dimensional search spaces based on active user state vectors.
- Designed **Content-Based Filtering (CBF)** layers using **cosine similarity** to rank high-dimensional product features.

Samsung Research Institute

Noida, India · Sep 2020 – Sep 2021

Software Engineer (CL 2)

- Architected a **continuous on-device perception pipeline** ingesting multi-modal media streams for automated ML classification.
- Designed a **cross-profile data brokering engine** leveraging multi-user storage isolation to route assets from **User 1** to sandboxed **User 0**.

Projects

JitterNot

Jan 2022 – Mar 2022

LSTM-Based Model Predictive Controller for Adaptive Video Streaming

UCSB

- Developed an LSTM network for real-time throughput prediction, transforming the closed-loop control system from MISO to SISO.
- Implemented an MPC controller leveraging Receding Horizon Control (RHC) principles to dynamically calculate optimal video resolution.

Autonomous Ground Vehicle & Swarm Robotics

Jan 2016 – Dec 2018

End-to-End Perception, Planning, and Multi-Agent Navigation Stack

IIT Kharagpur

- Developed a hardware-agnostic navigation stack in ROS, executing an Extended Kalman Filter (EKF) to fuse multi-modal sensor streams (LiDAR, IMU, GPS) for zero-shot **Sim2Real transfer**.
- Implemented real-time visual obstacle segmentation using a custom U-Net architecture alongside a hybrid motion planner combining global ($A^*/$ Dijkstra) paths with a Dynamic Window Approach (DWA) for local collision avoidance.
- Formulated a decentralized swarm coordination network utilizing peer-to-peer relative pose estimation via AprilTags over an ad-hoc XBee mesh topology.

In Motion — HAR using DeepConvLSTM

Nov 2019 – Nov 2019

Distributed Mobile Computing System for Human Activity Recognition

IIT Kharagpur

- Designed a client-server telemetry pipeline streaming high-frequency IMU sensor arrays for rolling window inference.
- Implemented a DeepConvLSTM core architecture using TimeDistributed CNN layers to model spatial-temporal features efficiently.

CohesiveAR

Apr 2022 – Jun 2022

Augmented Reality Pipeline for Texture Extraction and Homography Mapping

UCSB

- Developed a spatial perception pipeline using Google ARCore and OpenCV to map and project dynamic 3D scene geometry into unified 2D reference frames.
- Implemented matrix homography and perspective rectification algorithms to correct for camera tilt skew, extracting invariant surface textures from variable viewpoints.

Tweeter: Distributed Infrastructure Testbed

Sep 2021 – Dec 2021

High-Concurrency Microblogging Testbed on AWS EC2 Clusters

UCSB

- Conducted high-throughput load testing up to 512 users/sec, mapping user trajectories as state machines to latency scaling curves.

- Optimized read throughput by deploying nested fragment caches and database query preloading to eliminate bottleneck points.

Teaching

University of California, Santa Barbara

Graduate Teaching Assistant

Santa Barbara, CA

Sep 2021 – Jun 2023

- Directed instructional labs, formulated algorithmic programming assignments, and monitored system architecture milestones for senior capstone teams.
- Covered advanced and core tracks: CS189 (Capstone), CS184 (Android), CS24 (C++ Data Structures), and CS16 (Problem Solving).

Academic Service & Peer Review

Technical Program Committee & Peer Reviewer

ACM Symposium on User Interface Software and Technology (UIST)

May 2024

Served as an external peer reviewer evaluating full paper submissions for novel interface hardware, interactive system architectures, and spatial tracking methodologies.



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
STATEMENT OF GRADES OBTAINED FOR THE 8 SEMESTER COURSE IN ENGINEERING/TECHNOLOGY LEADING TO THE AWARD OF
BACHELOR OF TECHNOLOGY (HONOURS)



Roll No: 16EE10042

Name: SATYAM AWASTHI

Year of Admission : 2016-2017

Year of Graduation : 2019-2020

Course: B.Tech.(Hons.) in ELECTRICAL ENGINEERING

For Semester 1 SGPA: 8.09 CGPA: 8.09				
Subno	Name	L-T-P	CRD	GRD
CY11001	CHEMISTRY	3-1-0	4	B
CY19001	CHEMISTRY LAB.	0-0-3	2	EX
EA10001	EXTRA ACADEMIC ACTIVITY-I	0-0-3	0	B
EE11001	ELECTRICAL TECHNOLOGY	3-1-0	4	C
EE19001	ELECTRICAL TECHNOLOGY LAB.	0-0-3	2	B
HS13001	ENGLISH FOR COMMUNICATION	3-0-2	4	B
MA10001	MATHEMATICS-I	3-1-0	4	B
ME19001	INTRODUCTION TO MANUFACTURING PROCESSES	0-0-3	2	A

For Semester 2 SGPA: 8.91 CGPA: 8.51				
Subno	Name	L-T-P	CRD	GRD
CE13001	ENGINEERING DRAWING AND COMPUTER GRAPHICS	1-0-3	3	A
CS10001	PROGRAMMING AND DATA STRUCTURES	3-0-0	3	EX
CS19101	PROGRAMMING AND DATA STRUCTURES TUTORIAL AND LABORATORY	0-1-3	3	EX
EA10002	EXTRA ACADEMIC ACTIVITY-II	0-0-3	0	B
MA10002	MATHEMATICS-II	3-1-0	4	A
ME10001	MECHANICS	3-1-0	4	B
PH11001	PHYSICS	3-1-0	4	B
PH19001	PHYSICS LAB.	0-0-3	2	A

For Semester 3 SGPA: 8.68 CGPA: 8.57				
Subno	Name	L-T-P	CRD	GRD
EA10003	EXTRA ACADEMIC ACTIVITY-III	0-0-3	0	Y
EC21103	INTRODUCTION TO ELECTRONICS	3-1-0	4	B
EC29003	INTRODUCTION TO ELECTRONICS LAB.	0-0-3	2	B
EE21101	SIGNALS & NETWORKS	3-1-0	4	B
EE29001	SIGNALS & NETWORKS LAB.	0-0-3	2	A
HS30103	ECONOMICS OF HEALTH	3-0-0	3	EX
MA20101	TRANSFORM CALCULUS	3-0-0	3	A
PH20003	PHYSICS-II	3-1-0	4	A

For Semester 4 SGPA: 8.96 CGPA: 8.67				
Subno	Name	L-T-P	CRD	GRD
BS20001	SCIENCE OF LIVING SYSTEM	2-0-0	2	A
EA10004	EXTRA ACADEMIC ACTIVITY-IV	0-0-3	0	Y
EC21008	ANALOG ELECTRONIC CIRCUITS	3-1-0	4	A
EC29008	ANALOG CIRCUITS LAB.	0-0-3	2	EX
EE21002	ELECTRICAL MACHINES	3-1-0	4	A
EE21004	MEASUREMENTS AND ELECTRONIC INSTRUMENTS	3-1-0	4	A
EE29002	ELECTRICAL MACHINES LAB.	0-0-3	2	A
EE29004	MEASUREMENTS AND ELECTRONIC INSTRUMENTS LAB.	0-0-3	2	EX
EV20001	ENVIRONMENTAL SCIENCE	2-0-0	2	B
HS30074	SCIENCE AND HUMANISM	3-0-0	3	B

For Semester 5 SGPA: 9.04 CGPA: 8.75				
Subno	Name	L-T-P	CRD	GRD
CS40019	IMAGE PROCESSING	3-0-0	3	A
EC31003(*)	DIGITAL ELECTRONIC CIRCUITS	3-1-0	4	A
EC39003(*)	DIGITAL ELECTRONIC CIRCUITS LAB.	0-0-3	2	EX
EE31009	CONTROL SYSTEM ENGINEERING	3-1-0	4	A
EE31011	POWER ELECTRONICS	3-1-0	4	B
EE39006	POWER ELECTRONICS LAB.	0-0-3	2	EX
EE39009	CONTROL AND INSTRUMENTATION LAB.	0-0-3	2	B
EP60025	SPECIAL TOPICS IN ENTREPRENEURSHIP	2-1-0	3	EX

For Semester 6 SGPA: 9.46 CGPA: 8.87				
Subno	Name	L-T-P	CRD	GRD
CS31702(*)	COMPUTER ARCHITECTURE AND OPERATING SYSTEM	4-0-0	4	EX
CS60092	INFORMATION RETRIEVAL	3-0-0	3	B
EE30004	EMBEDDED SYSTEMS	3-0-0	3	EX
EE30012	ELECTROMAGNETIC THEORY & APPL.	3-0-0	3	A
EE31002	POWER SYSTEMS	3-1-0	4	A
EE39002	POWER SYSTEMS LAB.	0-0-3	2	EX
EE39004	EMBEDDED SYSTEMS LAB.	0-0-3	2	EX
MA20106	PROBABILITY & STOCHASTIC PROCESSES	3-0-0	3	EX

For Semester 7 SGPA: 9.25 CGPA: 8.92				
Subno	Name	L-T-P	CRD	GRD
CS60009(*)	SMARTPHONE COMPUTING & APPLICATION	3-0-0	3	EX
CS60021	SCALABLE DATA MINING	3-0-0	3	A
EE41001	INDUSTRIAL INSTRUMENTATION	3-1-0	4	B
EE47009	PROJECT-I	0-0-0	3	EX
EE48001	INDUSTRIAL TRAINING	0-0-0	2	A
EE49001	CONTROL AND ELECTRONIC SYSTEM DESIGN	0-0-3	2	A
EP60044	FRUGAL ENGINEERING	3-0-0	3	EX

For Semester 8 SGPA: 9.63 CGPA: 8.98				
Subno	Name	L-T-P	CRD	GRD
CS31006	COMPUTER NETWORKS	3-0-0	3	EX
EE40002	ELECTRIC DRIVES	3-0-0	3	B
EE47008	PROJECT - PART 2	0-0-0	6	EX
EE48002	COMPREHENSIVE VIVA VOCE	0-0-0	2	EX
EE49004	POWER APPARATUS & SYSTEM DESIGN	0-0-3	2	EX

Additional subjects taken into account for earning a Minor					
Subno	Name	L-T-P	CRD	Semno	GR
CS21003	ALGORITHMS - I	3-1-0	4	4	A
CS29003	ALGORITHMS LABORATORY	0-0-3	2	4	A
CS60050	MACHINE LEARNING	3-0-0	3	5	B
CS60118	CLOUD COMPUTING	3-0-0	3	8	EX

* sign against a major curricular subject indicates that it has been taken into account for Minor

GPA in Minor: 9.36

Minor in : COMPUTER SCIENCE AND ENGINEERING

Details of other additional subjects

Subno	Name	L-T-P	CRD	Semno	GRD
EC21107	SEMICONDUCTOR DEVICES	3-1-0	4	3	A

Total Additional Credit Taken: 16
GPA in Additional Subjects: 9.00

Total Additional Credit Cleared: 16

Total Credit Taken in Major Curriculum: 176 Total Credit Cleared: 176 CGPA: 8.98

Checked by

Joint Registrar (Academic)

Joint Registrar (Academic) / Assistant Registrar(UG)

Date of Issue : 16 Jul 2020

University of California, Santa Barbara

Office of the Registrar, Santa Barbara, CA 93106-2015

PARCHMENT: TWCTPVIN

PRINTED: 04/05/23

PAGE: 1

OFFICIAL TRANSCRIPT

STUDENT NAME: SATYAM AWASTHI

PERM NUMBER: 807498

SSN: XXX-XX-4913

Birth Date: 01/23/XX

CURRENT ACADEMIC PROGRAM:

COLLEGE: College of Engineering

OBJECTIVE: Master of Science

MAJOR: Computer Science

COURSE	TITLE	GRADE	COMPLETE UNITS	GPA UNITS	GRADE POINTS	GPA
2021 Fall Quarter						
Admitted into the Master of Science Program in Computer Science						
Admitted into the Master of Science Program in Computer Science						
CMPSC	291A	SPEC TOPIC APPS GEN	A+	4.0	4.0	16.00
CMPSC	501	TECH COMP SCI TEACH	S	1.0	0.0	0.00
CMPSC	502	TEACHING COMPTN SCI	S	3.0	0.0	0.00
LING	4	INDIV & GRP INSTRU	P	0.0	0.0	0.00
No Baccalaureate Credit Lower-Division Course						
Graduate Total:				8.0	4.0	16.00
THRU F21				8.0	4.0	16.00
2022 Winter Quarter						
CMPSC	271	ADV DIST SYS	A	4.0	4.0	16.00
CMPSC	291A	SPEC TOPIC APPS GEN	A	4.0	4.0	16.00
CMPSC	293C	SPEC TOP SYS PROG L	A	4.0	4.0	16.00
Graduate Total:				12.0	12.0	48.00
THRU W22				20.0	16.0	64.00
2022 Spring Quarter						
CMPSC	292C	SPEC TOP FOUN PROG	A+	4.0	4.0	16.00
CMPSC	293N	SPEC TOPICS CS SYS	A	4.0	4.0	16.00
CMPSC	595C	PRG LANG SOFT ENGR	S	2.0	0.0	0.00
Graduate Total:				10.0	8.0	32.00
THRU S22				30.0	24.0	96.00
2022 Summer Session						
CMPSC	193	INTERN IN INDUSTRY	P	1.0	0.0	0.00
Graduate Total:				1.0	0.0	0.00
THRU M22				31.0	24.0	96.00

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ISSUED TO:

SATYAM AWASTHI
6510 El Colegio Rd Apt 1102
Santa Barbara CA 93106-4200
Anthony Schmid
University Registrar

University of California, Santa Barbara

Office of the Registrar, Santa Barbara, CA 93106-2015

PARCHMENT: TWCTPV1N

PRINTED: 04/05/23

PAGE: 2

OFFICIAL TRANSCRIPT

STUDENT NAME: SATYAM AWASTHI

PERM NUMBER: 807498

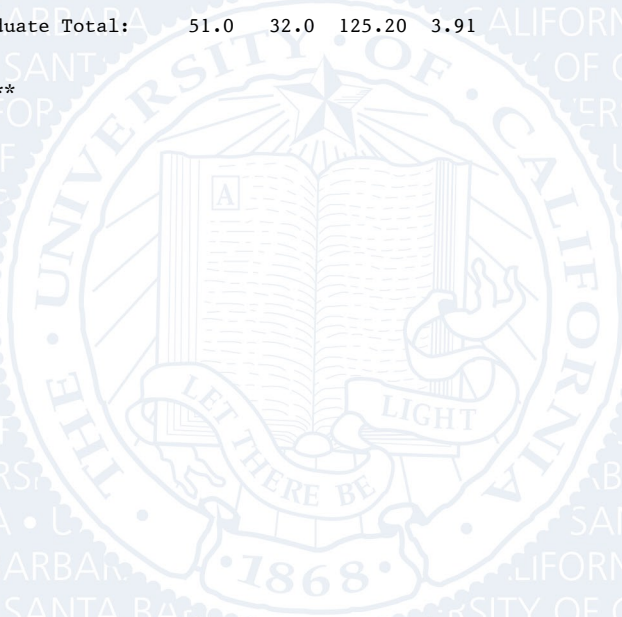
COURSE	TITLE	GRADE	COMPLETE GPA		GRADE POINTS		GPA
			UNITS	UNITS			

2022 Fall Quarter							
CMPSC	263	RUNTIME SYSTEMS	A	4.0	4.0	16.00	
CMPSC	291A	SPEC TOPIC APPS GEN	B+	4.0	4.0	13.20	
CMPSC	596	DIRECTED RESEARCH	S	4.0	0.0	0.00	
Graduate Total:				12.0	8.0	29.20	3.65
THRU F22	Graduate Total:			43.0	32.0	125.20	3.91

2023 Winter Quarter							
CMPSC	502	TEACHING COMPTR SCI	S	4.0	0.0	0.00	
CMPSC	596	DIRECTED RESEARCH	S	4.0	0.0	0.00	
Graduate Total:				8.0	0.0	0.00	0.00
THRU W23	Graduate Total:			51.0	32.0	125.20	3.91

UC Credit	Graduate Total:			51.0	32.0	125.20	3.91

** END OF RECORD **



ISSUED TO:

SATYAM AWASTHI
6510 El Colegio Rd Apt 1102
Santa Barbara CA 93106-4200



Anthony Schmid
University Registrar

UNIVERSITY OF CALIFORNIA • SANTA BARBARA

TRANSCRIPT GUIDE

UNITS OF CREDIT:

Until September 1966, credits were recorded as semester units. Since that time, the University of California, Santa Barbara has operated on a quarter calendar, with three quarters per academic year and an optional summer session. Each quarter has ten weeks of instruction at the rate of one unit for every three hours of student work required each week.

CURRENT GRADING SYSTEM:

The grades A, B, C, and D may be modified by plus (+) or minus (-) suffixes. Grade points for each unit are assigned as follows:

A+ = 4.0	A = 4.0	A- = 3.7
B+ = 3.3	B = 3.0	B- = 2.7
C+ = 2.3	C = 2.0	C- = 1.7
D+ = 1.3	D = 1.0	D- = 0.7

Grades of F, I, IP, P, NP, S, U and W = 0.0 grade points

Unit credit, but not grade-point credit, is assigned for P and S grades.

P = Passed (C or better)	Undergraduate Course Only
NP = Not Passed (C- or below)	Undergraduate Course Only
S = Satisfactory (B or better)	Graduate Course Only
U = Unsatisfactory (B- or below)	Graduate Course Only
I = Incomplete	
W = Withdrawal after established deadline	
IP = Grade to be assigned at the end of the final course in the sequence	

GOOD ACADEMIC STANDING:

A student is in good standing academically unless otherwise indicated.

TRANSFER CREDIT:

Only credit that is accepted by the University is shown on the transcripts of UCSB students. Credit from other UC campuses may be shown either by individual course records or as a unit summary by campus. Individual courses from schools outside of the UC system are not shown.

ADVANCED PLACEMENT CREDIT:

Examinations and the credits accepted for them are shown on the transcript in the same format and location as transfer credit.

COURSE NUMBERING SYSTEM:

1 - 99	Lower Division Courses
100 - 199	Upper Division Courses
200 - 299	Graduate Courses
300 - 399	Professional courses for teachers
400 & above	Other professional and graduate courses

REPEATED COURSES:

An undergraduate student may repeat only those courses for which an NP or letter grade of C- or lower is recorded. If the grade from a previous attempt had been included in the GPA calculations, its effect is then removed by a "repeat adjustment" up to a maximum of 16 units over the student's career.

ACCREDITATION:

Western Association of Schools and Colleges (WASC)